Lecture 10 — Clustering and k-means

Fri Sep 18

Last updated: September 13, 2026.

Boyd video (optional): Lecture 13 — k-means · Lecture 14 — k-means applications · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 4.1–4.3 (clustering objective and k-means). Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Explain the goal of clustering.
2. Compute one assignment/update iteration of k-means.
3. Interpret a centroid as a vector.
4. Recognize that clustering output depends on representation, scaling, and initialization.

Warm-up

Six points in $\mathbb{R}^2$:

$$(0,0),\ (0,1),\ (1,0),\quad (5,5),\ (5,6),\ (6,5).$$

By eye, into how many natural groups do they fall? Suggest a representative (center) for each group.

Fill in

Clustering partitions N vectors $x_1, \ldots, x_N$ into k groups so that vectors in the same group are _____ to each other (small pairwise distances).

A **group representative** (centroid, when we use the mean) z_j is a vector that stands for cluster j. The basic clustering objective sums squared distances from each point to its group's representative:

$$J^{\text{clust}} = \frac{1}{N}\sum_{i=1}^{N}\|x_i - z_{c_i}\|^2,$$

where c_i is the cluster index of x_i.

k-means alternates two steps:

1. **Assignment:** given representatives $z_1, \ldots, z_k$, set each c_i to the index of the _____ z_j to x_i.
2. **Update:** given assignments, replace each z_j by the _____ of the points currently in cluster j.

The algorithm is a _____: different initial z_j can yield different final clusterings. Feature _____ and the choice of k also change the result.

Worked in class

W1. Goal in words. A café has purchase vectors (coffee, tea, pastry counts) for many customers. What would clustering these vectors help the café do?

W2. One assignment step. Data:

$$x_1 = (0, 0),\ x_2 = (0, 2),\ x_3 = (8, 0),\ x_4 = (8, 2).$$

Representatives $z_1 = (1, 1)$, $z_2 = (7, 1)$. Assign each x_i to the nearest z_j. List the two clusters.

W3. One update step. Using your clusters from W2, replace each z_j by the mean of its points. What are the new centroids?

W4. Interpret a centroid. If a cluster of temperature vectors has centroid $(54, 53, 55)$ for three morning hours, what does that vector mean?

W5. Sensitivity. Same four points as W2, but start with $z_1 = (0, 1)$, $z_2 = (0, 0)$. After one assignment, what goes wrong if both points near the origin grab almost everything? (One sentence on initialization.)

Check yourself

C1. In k-means assignment, each point is assigned by a _____ computation to the current representatives.

C2. For points $(0, 0), (2, 0)$ in one cluster, the centroid is _____.

C3. True or false: running k-means once always finds the global minimum of J^{clust}.

C4. Why might you standardize features before clustering customer data that mixes age and annual spending?

C5. If $k = N$ (one cluster per point), what is J^{clust} after a perfect fit of representatives?

Answers (Check yourself)

C1. nearest-neighbor (distance to each representative)

C2. $(1, 0)$

C3. **False.** It is a heuristic; different initializations can give different local solutions.

C4. So spending in dollars does not dominate age in years inside Euclidean distance.

C5. $J^{\text{clust}} = 0$ (each point is its own representative).